

Revenue Recognition and Margins

Session 9 · Wednesday, September 30, 2026

Case 2 memo due Fri 10/2, 1:00 pm · PS2 (Redline Labs) due Mon 10/12, 1:00 pm

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Roadmap: the five questions of revenue

1. **What counts as revenue?** — bookings, billings, ARR vs. GAAP
2. **When do I recognize it?** — ASC 606's five steps
3. **What if cash comes first?** — deferred revenue (PS2's engine)
4. **Bundles & SaaS** — together or apart?
5. **Whose revenue is it — and what's left over?** — gross vs. net, then margins

Four 'you make the call' rounds — commit BEFORE the rule. • Case 2 memo due Fri 10/2 • PS2 (Redline Labs) due Mon 10/12, 1:00 pm.

What is "revenue"? Two IPOs and one statistic

Company	What they do	The surprise
Rubicon Project (NYSE: RUBI, IPO Feb 2014)	Automates buying/selling of digital advertising; paid a fee per transaction.	Touted growth as managed revenue — the gross ad dollars flowing through, not its per-transaction fees. Post-IPO investors felt misled.
Hortonworks (NASDAQ: HDP, IPO Dec 2014)	Platform to store, manage, analyze data.	CEO guided bankers on billings, a non-GAAP gauge of future business. Hortonworks and its VC (Benchmark) thought they were transparent; the bankers disagreed.

Andreessen Horowitz: for startups, traditional GAAP metrics 'can lead to bad investment decisions' — non-GAAP measures can better predict prospects.

61% of SEC accounting-related enforcement actions are revenue-recognition issues. (COSO — Committee of Sponsoring Organizations of the Treadway Commission)

The CFO must speak BOTH dialects — GAAP and non-GAAP — and label them honestly.

Bookings vs. revenue vs. cash ★



Booking — customer commits via a contract to buy. (Signed ≠ earned.)

+ **Delivery** — the product/service is actually delivered.

+ **Reasonable assurance of collection** — cash will be collected → **Revenue (GAAP)**.

Five situations — each resolves on a later slide:

eCommerce: gross vs. net → #9 · Paid up front / deferred → #19 · Software invoiced for future payment → #8

Hardware + software bundle → #24 · Signed but not yet implemented → #14

The bookings metrics zoo: MRR, ARR, ACV, TCV ★

Metric	Definition	What it's used for
New Accounts Added	count of new accounts in a period	customer growth; S&M efficiency
MRR (Monthly Recurring Revenue)	recurring subscription value per month; excludes one-time fees	revenue momentum, user growth, churn/retention
ARR (Annual Recurring Revenue)	recurring subscription value annualized; excludes one-time fees	the valuation metric for software
ACV (Annual Contract Value)	annual contract value; includes one-time fees	pairs with CAC to compute payback period
TCV (Total Contract Value)	whole-contract value over its life; includes one-time + recurring	growth management, expenditure planning

Cold-call: 1. What does a high ARR:ACV ratio show? 2. What does a high TCV:ARR ratio show?

Exercise: compute the four metrics

Pairs — 3 minutes. Two customers each sign a 24-month contract in month 1. Both pay a one-time activation fee. Customer A and Customer B pay different monthly fees.

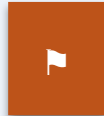
Compute: (1) MRR, (2) ARR, (3) ACV Year 1 and Year 2, (4) TCV.

Customer A



Recurring fee \$100/mo — months 1-24

Customer B



Recurring fee \$200/mo — months 1-24

▀ One-time activation fee at month 1 (both customers) · blue bar = recurring monthly fee

Deliverable: four numbers per pair.

activation \$400 each · A \$100/mo · B \$200/mo

Billings = Revenue + Δ Deferred Revenue ★

$$\text{Billings} = \text{Revenue} + \Delta \text{ Deferred Revenue}$$

Billings = (i) sales to new customers + (ii) renewals + (iii) expansion within existing customers.
= whatever is invoiced in the period — even a multi-year invoice.

"We typically invoice customers at the beginning of the term — multiyear, annual, quarterly, or monthly... we do not consider [billings] a non-GAAP measure given it is calculated exclusively from revenue and deferred revenue, both GAAP measures."

Box Inc. 2017 10-K.

You make the call #1: Groupon — how much revenue?

2011. A customer pays Groupon \$40 for a restaurant voucher. Groupon keeps \$20 and remits \$20 to the restaurant. How much revenue does Groupon report?

- A. \$40 — Groupon sold the voucher
- B. \$20 — Groupon earned a fee
- C. \$0 until the voucher is redeemed
- D. \$40 revenue, \$20 marketing expense

Commit — hands on my count.

ASC 606: five questions, five steps ★

Revenue is recognized when it is **earned and realizable** — ASC 606 operationalizes this in five steps.



1. Contract — Do I have a contract with my customer?
2. Products & Services — What product(s)/service(s) am I obligated to deliver?
3. Transaction Price — How much revenue will I earn for delivering them?
4. Allocating Revenue — If I deliver multiple products/services, how should I allocate revenue?
5. Delivery — When have I completed my obligations and earned the revenue?

'Revenue from Contracts with Customers' (ASC 606) — effective for periods after Dec 15, 2017.

Step 1 — Is there a contract? (and will they pay?) ★

1

2

3

4

5

Five contract criteria:

- Approval by both parties
- Rights of each party identified
- Payment terms defined
- Has commercial substance
- **Collectability is probable (Merck)**

Assess: knowledge of the customer (payment history / credit rating); ability to stop service on non-payment.

Selling to a start-up? Or an at-risk retailer (e.g., Sears)? What do you book?

No written contract? (Startup reality)

1

2

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No contract at inception → recognize revenue only when:

- substantially all consideration is received and nonrefundable, OR
- the contract is terminated and consideration nonrefundable, OR
- control has transferred, performance has stopped with no remaining obligation, and consideration is nonrefundable.

2ndKitchen — early startups sell without paper to cut friction; the accountant's job is to know what that defers.

Step 2 — What am I obligated to deliver? ★

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Two "distinct" tests:

- Customer can benefit from it on its own (or with readily available resources)
- Promise is separately identifiable from others in the contract

Three NOT-separately-identifiable flags:

- Significant integration service
- Significant modification / customization
- Highly interdependent / interrelated

If not separately identifiable → recognize revenue only when ALL obligations are delivered.
(MicroStrategy)

SaaS implementation = usually distinct; heavily customized software = not distinct.

This one paragraph decides Fitbit, Recruit Right, and Precedent — slides 24-28.

Step 3 — How much will I earn? ★

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Transaction price = expected consideration for delivering the goods/services, **excluding amounts collected on behalf of third parties (AOL).**

Four adjustment factors:

- Variable consideration (discounts, concessions)
- Significant financing component
- Noncash consideration
- Amounts payable to the customer

Money comes in from a 3rd party and looks like revenue to an accountant who doesn't know it goes right back out.

Variable consideration — and Outcome Health ★

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Types: price concessions • volume discounts • rebates • refunds & credits • performance bonuses • royalties.

Two estimation methods: **Expected value** (probability-weighted range) vs. **Most likely amount** (single most-likely figure).

Outcome Health

Chicago health-media startup; \$5.5B valuation round, early 2017. Former executives accused it of overcharging for ads that under-delivered on screen counts. Investors sued for fraud tied to overstated revenue. The 'make-goods' / 'bonus-media' concessions were variable consideration — revenue that should never have been recognized.

Variable consideration takes judgment. In private companies, investors sue over that judgment.

Step 4 — Allocate the price across the bundle ★

1

2

3

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Allocate by **relative standalone selling prices**. Best evidence = the price when sold separately; otherwise management estimates.

Element	Price sold separately	Relative %	Allocated to \$1,200
Component 1	\$1,000	75%	\$900
Component 2	\$80	6%	\$72
Component 3	\$250	19%	\$228
Total	\$1,330	100%	\$1,200

Bundle priced at \$1,200; prices are not always easily separable. (MicroStrategy)

Step 5 — When? Point in time vs. over time ★

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Three over-time tests (any one → recognize over time):

- Customer simultaneously receives and consumes the benefit
- Performance creates/enhances an asset the customer controls
- Performance creates no alternative-use asset AND the company has an enforceable right to payment for work done to date

If none hold → recognize at a point in time.

"Booking revenues, like comedy, is all about timing." — The Economist

For long builds (airplanes, real estate), completed-contract would show no periodic progress; percentage-of-completion recognizes by cost-progress instead. Worked data-center example → appendix A4.

Deferred revenue: cash first, revenue later ★

Deferred (unearned) revenue = a **liability** recorded when cash is received before the revenue is earned (delivery of product/service).

Moment	Journal entry
At prepayment	Dr Cash / Cr Deferred Revenue
As earned	Dr Deferred Revenue / Cr Revenue

It is a **LIABILITY** — an obligation to **DELIVER**, not money owed to you.

The single most important balance-sheet line for SaaS — and the engine inside PS2's Redline Labs model.

You make the call #2: MedVR — 2018 revenue?

On 11/1/18, MedVR signs a 1-year services contract with Diamond Health for \$6,500,000, paid in full in cash on day one. By 12/31/18, MedVR has delivered 2 of 12 months. How much 2018 revenue?

- A. \$6,500,000 — cash is in the bank
- B. \$0 — the contract isn't done
- C. \$1,083,333
- D. \$541,667

And: what does the balance sheet show on 12/31?

Why software revenue is the regulators' favorite

Three complexity sources:

- Does a bundled arrangement exist?
- Timing for a bundle (hardware+software OR software+services)?
- How to assess a SaaS arrangement?

These complexities are why FASB issued ASC 606.

License vs. SaaS: product vs. service ★

Software as a product

- License = the deliverable
- Customer or a 3rd party hosts
- Revenue at delivery, minus deferred support

Microsoft: most revenue up front, support deferred.

SaaS

- Right to access; no ownership transfers
- Vendor hosts
- Pure service → revenue ratable over the service period

Walk-through example (BizInsights) → appendix A2. The parallel capitalization asymmetry is in appendix A9–A10.

You make the call #3: Fitbit — the Day-1 entry?

A customer buys a Fitbit Blaze for \$150 cash on Jan 1. The required software/app service = 20% of the value; estimated use period = 24 months.

What does Fitbit's Day-1 journal entry credit?

- A. Revenue \$150
- B. Revenue \$120 + Deferred revenue \$30
- C. Deferred revenue \$150
- D. Revenue \$30 + Deferred revenue \$120

The bundling question: distinct, or one promise? ★

Licenses ship bundled with hardware, upgrades, post-contract support, services. Recognize separately only if the software's value stands without them — this requires judgment.

Distinct

→ allocate & recognize per element (Step 4 clocks).

Not distinct

→ one performance obligation; recognize when ALL delivered.

Two live rounds: Recruit Right (next) and Precedent (appendix).

You make the call #4: Recruit Right — one bundle or three?

6/30/18: Recruit Right signs a contract with a mid-size customer, \$4.5M paid in full. Perpetual software licenses. Implementation with no significant customization — installable by other vendors. 2-year customer support that internal IT could substitute.

Standalone-price estimates: licenses 65%, implementation 2%, support 33%.

One bundle, or three obligations?

- A. One bundle — recognize \$4.5M ratably over the 2-year support term
- B. Three distinct obligations — license at delivery; implementation as performed; support over 24 months
- C. All \$4.5M at license delivery — they paid up front
- D. License now; implementation + support only when the contract ends

Then compute: how many dollars land on each element? (65% / 2% / 33% of \$4.5M)

Gross vs. net: whose revenue is it? ★

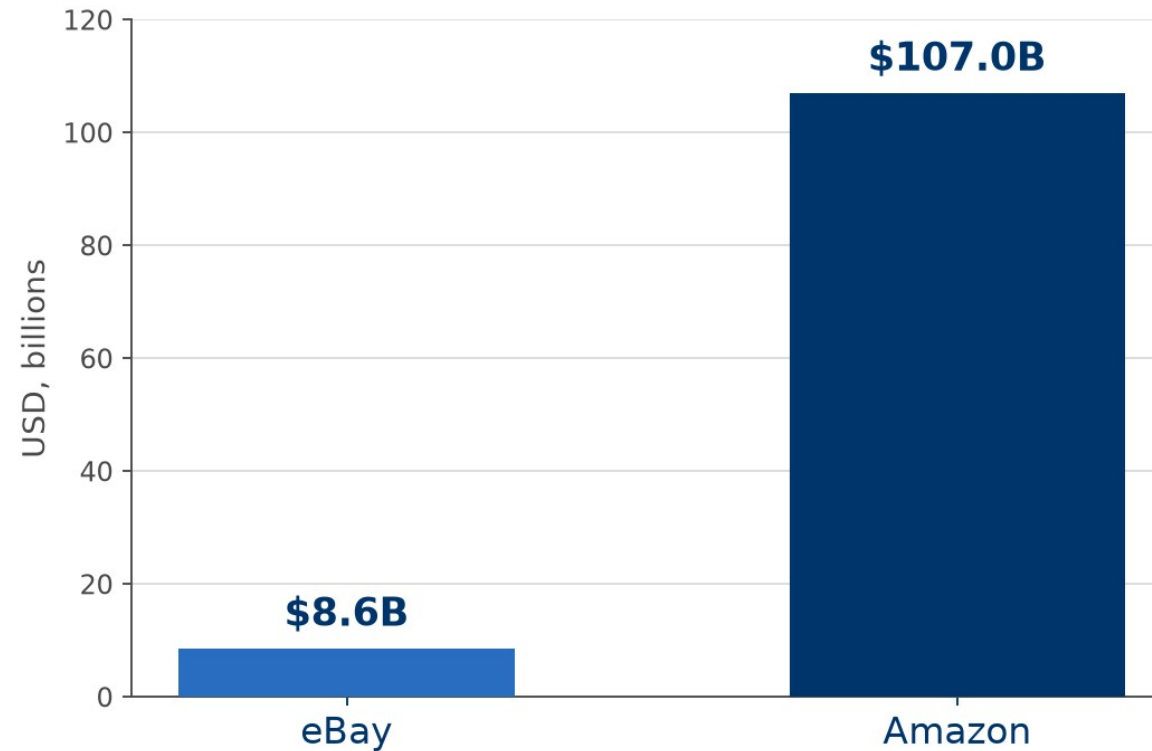
Principal (owns inventory, sets price, bears fulfillment/credit risk) → reports **GROSS**.

Agent (earns a fee for arranging) → reports **NET** = purchase price less amounts payable to the merchant.

GMV = total merchandise value transacted — a useful KPI to assess marketplace growth; typically NOT revenue.

This is the rule Groupon got wrong (slide 10).

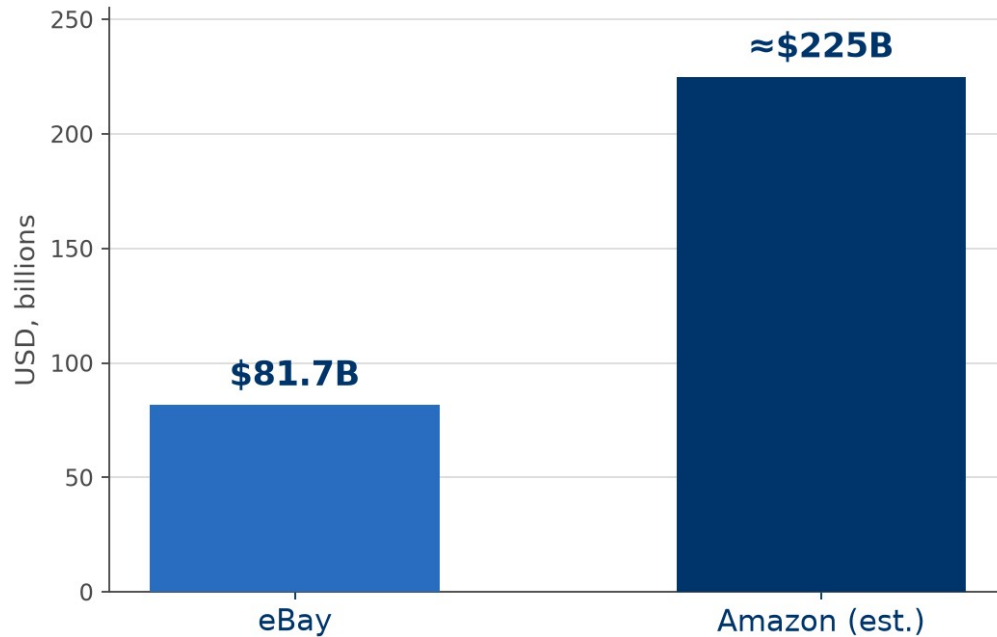
Amazon vs. eBay — the revenue view



Source: Amazon 2015 Form 10-K; eBay FY2015 results.

By revenue, eBay looks like a marginal player next to Amazon... is that the right comparison?

Amazon vs. eBay — the GMV view ★



Source: eBay FY2015 results (GMV); Amazon GMV = analyst estimates (not disclosed).

Amazon **retail** owns inventory → GMV counts as revenue.

Amazon **marketplace** → ~8% of GMV booked as service revenue.

eBay = pure marketplace → ~3% take rate (≈2.98%) booked as revenue.

Presentation changes revenue and margin percentages — not gross-profit dollars. Investors re-derive GMV anyway; CFOs who blur the line get the Groupon treatment.

Gross margin — and what belongs in cost of revenue ★

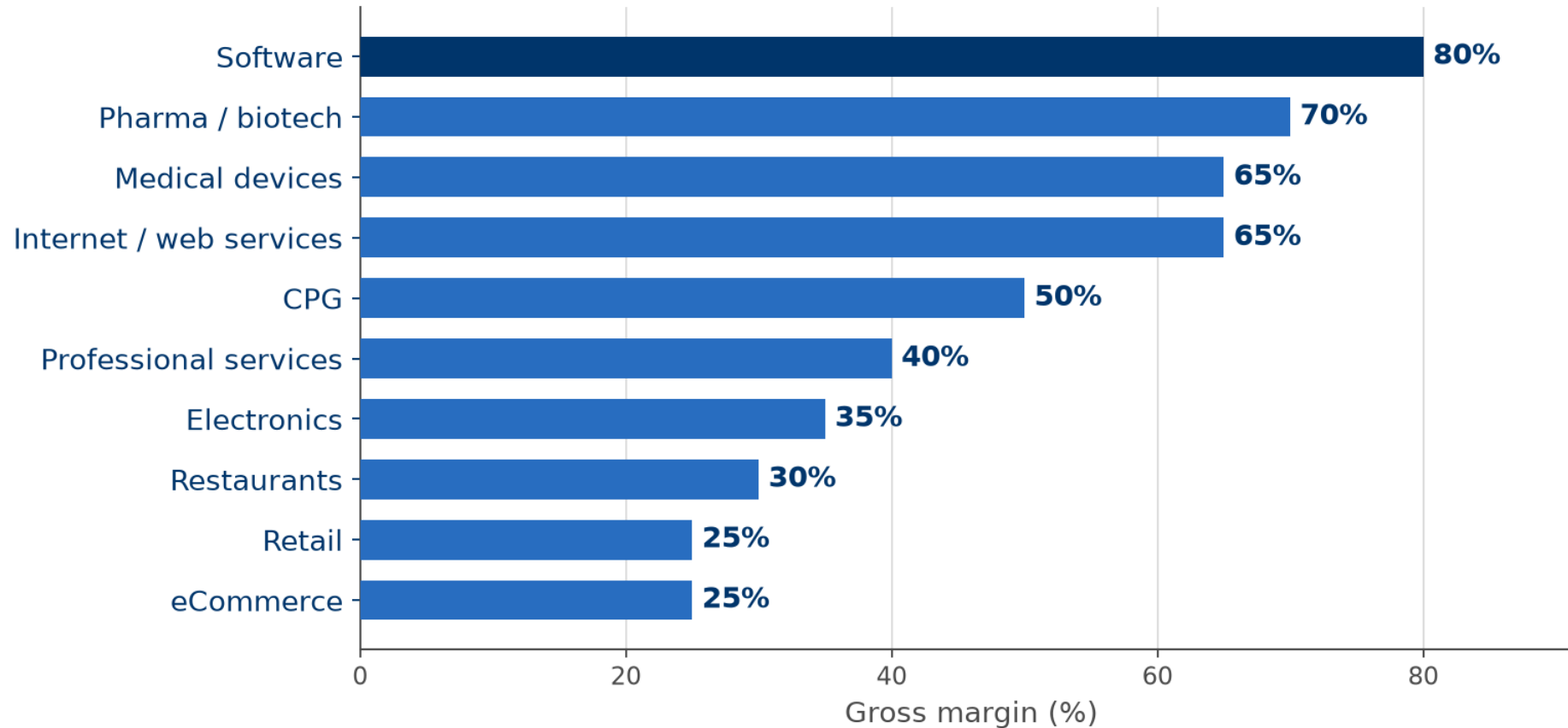
Gross Margin = (Revenue – Cost of Revenue) / Revenue. The formula is easy; the hard part is which costs go in.

IN cost of revenue	OUT (below the line)
Raw materials / merchandise	S&M salaries
Production & service-delivery labor	Advertising & marketing
Related overhead	Professional-services costs
	Travel & entertainment
	Corporate IT / infrastructure

Matching principle: expenses are recognized in the period of the related revenue. Also called COGS / cost of merchandise / cost of service / cost of sales.

Margin benchmarks: software $\approx$ 80%

Software $\approx$ 80% — this is WHY software multiples exist.



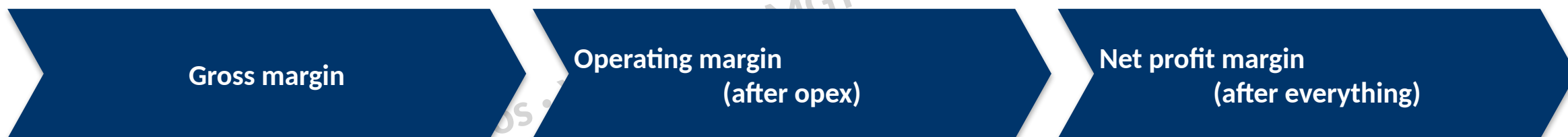
Source: review of 61 high-tech + 50 other public companies (illustrative ladder; software $\approx$ 80% is source-firm).

Cold-call: what's IN the SaaS cost line? $\rightarrow$ hosting, support personnel, capitalized-software amortization (appendix A11).

Contribution margin vs. gross margin — and the ladder ★

Contribution margin = per-unit profit lens. Excludes fixed costs; includes ALL variable costs — even variable S&M that sits below the gross-profit line.

Fixed costs are OUT of CM. GM can still carry fixed production overhead — so CM $\neq$ GM.



Unit price – variable cost = contribution / unit; CM ratio = contribution / price.

CM feeds the S06 unit-economics stack (CAC payback, LTV).

Sales commissions: capitalize, then amortize ★

Historically, commissions were expensed in the quarter of sale → poor matching on multi-year contracts.

Topic 606 (periods after Dec 15, 2017): capitalize sales commissions as incremental costs of obtaining a contract; expense as goods/services are delivered — **amortize over the benefit period.**

Consequence: a '**deferred commissions**' asset mirrors the deferred-revenue liability.

Mini-poll: Your AE closes a multi-year SaaS contract in January and is paid the full commission (\$60,000) in cash in January. How much commission expense hits THIS year's P&L? (3-year contract)

- A. the full commission (pre-2018 habit) B. the amortized slice C. \$0 — it's an asset forever D. depends on collecting the receivable

Takeaways ★

1. Bookings sign, billings invoice, revenue delivers — never present one as another. ★
2. ASC 606: contract → obligations → price → allocate → recognize; collectability gates the door. ★
3. Cash before performance = deferred revenue — a liability, and SaaS's growth gauge. ★
4. Gross vs. net is principal vs. agent; GMV is a KPI, not revenue. ★
5. Margins: gross ≠ contribution; commissions now capitalize and amortize (Topic 606). ★

PS2 (Redline Labs) due Mon 10/12, 1:00 pm — annual-prepay deferred revenue is the model's engine (slide 21).

Monday: Case 2 discussion (Solstice) — memo due Fri 10/2. · Office hours Tue 10/6, 1:30–3:00. · Wed 10/7: board packages — bring today's toolkit.

Appendix — reference only, examinable at concept level.

Worked extras: BizInsights (SaaS), Precedent (bundle that fails 'distinct'), percentage-of-completion, Amazon's five revenue models, dev-cost capitalization, SaaS cost of revenue.

Appendix: SaaS walk-through — BizInsights

BizInsights hosts a business-analytics platform accessed via the cloud; the customer cannot take control of the platform; no on-premise version; no 3rd party can host it.

Q: Does the contract transfer a software license?

No — the customer can't control or host it.

Q: Mixed point-in-time + over-time?

No — no license element; recognize **ratably over the service life**.

Appendix: Bundled software example 2 — Precedent

Precedent sells legal-precedent software (\$2M perpetual license) + **daily AI case-law updates (\$8M over 5 years)**. Without the updates, the software risks outdated precedent — value significantly impaired.

Updates are NOT separately identifiable → single performance obligation → recognize \$10M over 5 years.

Same two-element shape as Recruit Right (#28), opposite answer — the facts decide.

Appendix: Percentage-of-completion — Data Centric

Beginning 2019, Data Centric contracts to build a large-scale data center for \$5M, ~3 years, with a prearranged installment schedule (customer mines crypto).

Recognize by **cost-progress ratio** (costs incurred ÷ total estimated cost), not at final delivery.

Caveat: estimated cost-to-complete changes over time.

Appendix: Amazon = many revenue models

Amazon is retailer AND marketplace — and more. Distinct revenue-recognition policy per unit:



Appendix: Revenue-model policies I — retail & marketplace

Company / BU	Description	Revenue-recognition policy
Amazon Retail	owns inventory, sells directly	recognized when products ship and title passes to customers.
Blue Nile	online jewelry; taken private by Bain, Feb 2017	recognized on the date ownership passes, net of estimated returns and promotional discounts.
Amazon Marketplace	3rd-party sellers; GMV > retail revenue	net commissions recorded when not primarily obligated and no price latitude (fixed fees / % of seller revenue / per-unit fees).
Etsy	IPO 2015, NASDAQ: ETSY	3.5% transaction fee recognized at the transaction; \$0.20 listing fee recognized ratably over a 4-month listing period.

Appendix: Revenue-model policies II — devices, subscription, cloud

Company / BU	Revenue-recognition policy
Kindle / GoPro (multiple-element; GoPro: NASDAQ: GPRO)	device (substantial portion) recognized at delivery; undelivered software upgrades / support deferred and recognized straight-line over the estimated service period.
Amazon Prime / Wix (Prime \$119/yr, \$59 students; Wix IPO 2013, NASDAQ: WIX)	membership/subscription revenue amortized over the life of the membership based on estimated delivery of services / ratably over the service term.
AWS / cloud	service revenue recognized when the service is rendered.

Appendix: Costs before revenue — where do they park?

Matching → costs incurred before the related revenue are **capitalized as assets** (inventory; software development), not expensed yet.

$$\text{Beginning inventory} + \text{Purchases} - \text{Ending inventory} = \text{COGS}$$

Merchandising inventory = purchase cost + costs to get goods sale-ready.

Manufacturing inventory = raw materials + direct labor + manufacturing overhead.

Stryker 10-K: cost of sales comprises direct materials/supplies, manufacturing labor, depreciation and overhead to convert materials into finished product, plus distribution/freight/warehousing.

Appendix: Capitalizing software dev costs — product vs. SaaS ★ (concept)

Same software, different revenue model → different assets and operating income.

Product timeline

Expense R&D until technological feasibility → capitalize until general release → then amortize.

Microsoft: costs are R&D expense until technological feasibility; once reached, capitalized. (10-K FY ended 6/30/2016.)

SaaS / internal-use timeline

Expense pre-development → capitalize from application-development start → amortize after general release.

Salesforce: capitalizes cloud/internal-use costs during the application-development stage; preliminary and post-implementation activities expensed. (10-K FY ended 1/31/2017.)

Appendix: Dev-cost capitalization — impact & amortization

\$5M 2018 comparison: Product → dev costs expensed → lower NI, lower assets. SaaS → capitalized → NI unaffected, assets unchanged.

Amortization rule: each year amortize the **greater of** (a) current-product-revenue ÷ total-expected-product-revenue × capitalized cost, or (b) straight-line over remaining economic life.

Worked: \$25M capitalized; Year-1 revenue \$9M of \$30M expected; 4-year life → amortize \$7.5M = $\max((9/30) \times 25 = 7.5, 25/4 = 6.25)$.

Appendix: SaaS cost of revenue (what's in the 80%)

SaaS cost of revenue = **data-center / hosting capacity • support personnel • professional-services staff • amortization of capitalized software.**

Workday 10-K: cost of subscription revenues consists of third-party data centers, hosting/support personnel, computer-equipment leases, allocated overhead, and amortization of capitalized internal-use software. (10-K FY ended 1/31/2017.)